

movies slated for release in 2011. A Samsung spokesperson says that by 2014, 20 per cent of the Hollywood movies will be 3D. James Cameron is planning to make a 3D film on the Amazon Tribe, while Peter Jackson is all set to shoot *The Hobbit* in 3D. Plus, you could pick up video games for 3D TVs, disks of older 3D movies or even direct-to-video 3D movies that are likely to come out soon. Direct-to-video movies are essentially films made by smaller production houses that do not have budgets large enough for a theatre release. These movies are likely to be smaller, simpler and made especially for home viewing.

Channels. Disk-based content is certainly not enough to sustain a television technology. There needs to be a bit more action on the TV channels front too. In the United States, DirecTV has teamed up with 3D players such as Panasonic to offer channels like DirecTV Cinema. On the sports front, there is ESPN 3D. Discovery 3D and Sky Channel 3D are the other early entrants. Channels in countries like Japan and South Korea offer 3D-programmed content for a few hours every day. With a 3D-capable TV and set-top box, you could watch these channels if available in your region.

Web-based content. Google TV, when ready, is also expected to carry some 3D content. Samsung has announced an on-line 3D streaming facility. These channels are in a way fostering the anytime-anywhere culture as far as 3D television viewing goes. Since most 3D-ready televisions now have Ethernet and even wireless capabilities, you could sync up to one of these Web-based 3D channels. Plus, more and more 3D videos are being uploaded to YouTube. Moreover, television makers are also including easy-to-use applications and widgets to make finding and streaming of Web-based 3D content a no-brainer.

Certainly, options galore, but the fact is that the actual amount of 3D content is meagre compared to high-definition 2D, which itself is yet to attain full momentum in India. Inevitably, the availability of 3D content will increase, but owing to the high cost of production, it might turn out to be a catch-22 situation with TV makers waiting for content to increase, and content-makers waiting for TV costs and sales to stabilise.

Content through synergy

Brands such as Samsung, Sony and Panasonic are attempting to complement their products with a whole ecosystem consisting of content and applications, which will engage users still further. These are teaming up with movie-makers to make high-class 3D films, for launch not only in theatres but also in disk formats for home viewing. They are also sponsoring work on 3D channels.

Such a synergy between TV-makers and content-makers is definitely essential to promote 3D technology, because it is an expensive proposition. Creating content for 3D is 50 per cent

more expensive than regular films and it involves much more expensive cameras and post-production equipment. Hence film-makers will not venture into it, unless they are assured of a market and support system.

Indeed, TV manufacturers are teaming with the movie producers to make 3D films to woo the audience. Examples include Samsung's collaboration with DreamWorks Animation to make *Monsters vs Aliens*, *How to Train Your Dragon* and 3D versions of *Shrek*; Panasonic's collaboration with Twentieth Century Fox for bringing out *Coraline* and *Ice Age 3*; and Sony's release of *Cloudy with a Chance of Meatballs*.

While most of the recent 3D movies are computer-animated, only some such as *Coraline* and *Avatar* have been actually filmed with cameras—and the hardcore 3D effect is obvious in such movies, compared to the less genuine computer-made ones. Another issue is that most of these movies are reserved for a certain TV brand. That is, a TV company collaborates with a movie-maker, releases 3D Blu-ray or DVD versions, but gets exclusive rights to bundle it with its products, at least until its initial marketing spree ends. While this gains publicity for a television brand, it ensures a decent and assured return on the movie-makers' investment as well. For example, *Avatar* and *Coraline* are bundled with Panasonic TVs, while *Monsters vs Aliens* goes with Samsung TVs. *Cloudy with a Chance of Meatballs* is probably the only movie that is not brand-exclusive at the moment.

2D-to-3D conversion

There are hardware and software components available today that can be embedded in disk-players, set-top boxes and even televisions themselves, to enable the conversion of 2D content into 3D. It works by adding a dimension of depth to the content, doing the necessary splitting of images into separate streams for each eye, etc. However, it is

